

Benjamin Goldstein

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Portfolio: <https://bennyg9.github.io/>

EDUCATION

Georgia Institute of Technology College of Engineering

Bachelor of Science in Electrical Engineering | GPA: 4.0

Atlanta, GA

May 2029

Agoura High School (AHS)

Valedictorian | GPA: 4.4

Agoura Hills, CA

June 2025

SKILLS

Hardware: Arduino, ESP32, Raspberry Pi, Stepper/DC/Servo Motors, Sensors, Soldering, Multimeter, Oscilloscope

Software: Python, C++, C, Java, MATLAB, RISC-V Assembly, ROS2, Linux, Git, UART, VS Code, Onshape CAD, KiCAD

Libraries: Numpy, Matplotlib

Concepts: Embedded Systems, Motor Control (Stepper, PID), Sensing, DSP, Heuristic Search, ML Classifiers

PROJECT EXPERIENCE

Autonomous Rubik's Cube-Solving Robot

Agoura Hills, CA

Independent Project

Aug 2024 - Aug 2025

- Designed, fabricated, and integrated all mechanical, electronic, and software systems achieving <1 min autonomous solves
- Drove NEMA stepper motors with hall sensor feedback to ensure fast, accurate, and repeatable cube turning
- Built statistical color recognition algorithm using chi-square tests to reliably map cube faces under varied lighting conditions
- Implemented heuristic-driven depth-first search algorithm in Python on Raspberry Pi finding ≤ 35 move solutions

Autonomous Connect 4 Robot

Atlanta, GA

Independent Project

Sept 2025 - May 2026

- Developed custom IR detection circuitry with digital EMA and first difference signal filtering for reliable move detection analyzed graphically in MATLAB
- Implemented PID-controlled piece placement and optimized bitboard game representation with finite-state control logic
- Designed custom UART protocol enabling reliable bidirectional communication between Raspberry Pi and Arduino
- Programmed pruned minimax-based AI engine to achieve optimal gameplay decision-making at an 8-move search depth

Selected Hardware and Software Projects

Independent Projects

- Built a 9W spark-gap Tesla coil using flyback transformer circuitry, custom capacitor bank, generating 1-1.5 cm plasma arcs
- Trained CNN skin lesion classifier on NVIDIA Jetson Nano, detecting malignant lesions with ~90% accuracy
- Developed ML mini-projects including linear regression fuel efficiency model, logistic heart disease predictor, and neural network Flappy Bird player
- Designed and prototyped a tunable AM radio receiver using LC resonance, envelope detection, and analog audio amplification.
- Programmed real-time embedded side-scroller game on ESP32 in C with custom hash tables, dynamic memory management, and procedural map generation.

ADDITIONAL EXPERIENCE

Autobots VIP (Vertically Integrated Project)

Atlanta, GA

VIP Team Member

Jan 2026 - May 2026

- Developed and debugged ROS2-based mobile robotics stack for Turtlebot2 and Kobuki platforms, integrating base drivers, LiDAR, and odometry across Gazebo simulation and hardware testing
- Implemented custom ROS2 control nodes in Python for Gazebo simulation, processing LiDAR and odometry data to enable reactive navigation behaviors (obstacle avoidance and PID motion control).

Medical Robotics Club

Atlanta, GA

Limbo Electrical Control Team Member

Sept 2025 - Jan 2026

- Contributed to PID control system design driving fingers, thumb, and wrist actuation in a multi-DOF robotic prosthetic hand

RELEVANT COURSEWORK

Intro Signal Processing, Hardware/Software System Programming, Digital System Design, Differential Equations, Honors Linear Algebra, Calculus I/II, Physics (Mechanics, E&M, Modern), Object Oriented Programming, Statistics